



NeuroSpace

Complete evaluation export

Job ID	9b732709-4fe2-477e-a781-50f6f25f4f3e
Status	completed
Result timestamp	2026-06-01T18:56:04.392139
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Pipeline Configuration

Parameter	Value
model_types	rnn, qrnn
comparison_mode	pca
feature_selector	pca
manual_overrides.rnn_epochs	20
manual_overrides.qrnn_epochs	10
manual_overrides.batch_size	32
manual_overrides.rnn_batch_size	32
manual_overrides.qrnn_batch_size	16
num_classes	2
class_names	0, 1
qrnn_backend	qiskit
noise_enabled	True
noise_level	0.0010
mitigation_enabled	True
mitigation_runs	3

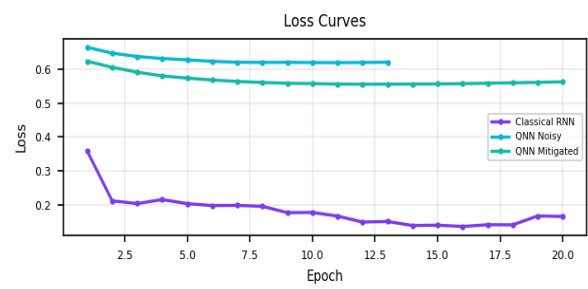
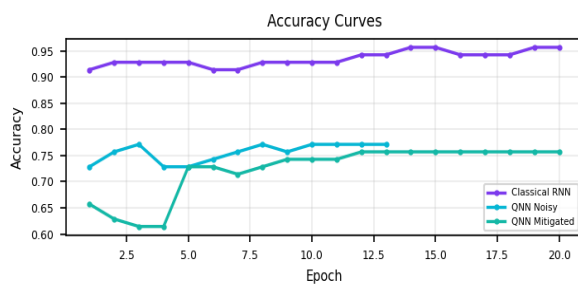
Best Model

Winner	RNN Accuracy	QNN Accuracy	Improvement
RNN	95.71%	82.14%	13.57%

Evaluation Summary

Model	Accuracy	Precision	Recall	F1 Score	Loss	Training Time
Classical RNN	95.71%	95.80%	95.71%	95.73%	0.1665	0.80s
QNN Noisy	76.43%	75.91%	76.43%	75.13%	0.6202	34.39s
QNN Mitigated	82.14%	82.03%	82.14%	81.49%	0.5624	2m 26.63s

Learning Curves



Classical RNN

Metric	Value
Accuracy	95.71%
Precision	95.80%
Recall	95.71%
F1 Score	95.73%
Loss	0.1665
Training Time	0.80s
Train Loss	0.0582
Val Loss	0.1665
Train Accuracy	98.36%
Val Accuracy	95.71%
Train Size	489
Val Size	70
Test Size	140

Classical RNN Confusion Matrix

Actual / Predicted	Class 0	Class 1
Class 0	88	4
Class 1	2	46

Classical RNN Classification Report

Class	Precision	Recall	F1 Score	Support
0	97.78%	95.65%	96.70%	92.0000
1	92.00%	95.83%	93.88%	48.0000
macro avg	94.89%	95.74%	95.29%	140.0000
weighted avg	95.80%	95.71%	95.73%	140.0000
Accuracy			95.71%	

Classical RNN Epoch History

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
1	0.4734	0.3592	76.48%	91.43%
2	0.1442	0.2126	97.14%	92.86%
3	0.1169	0.2045	97.34%	92.86%
4	0.0927	0.2162	97.75%	92.86%
5	0.1195	0.2043	96.11%	92.86%
6	0.1021	0.1986	96.93%	91.43%

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
7	0.0799	0.1991	97.55%	91.43%
8	0.0976	0.1964	96.52%	92.86%
9	0.0706	0.1777	97.96%	92.86%
10	0.0782	0.1783	97.55%	92.86%
11	0.0685	0.1677	97.75%	92.86%
12	0.0743	0.1501	97.96%	94.29%
13	0.0639	0.1518	98.36%	94.29%
14	0.0605	0.1399	98.36%	95.71%
15	0.0675	0.1408	97.96%	95.71%
16	0.0647	0.1371	97.96%	94.29%
17	0.0602	0.1422	97.75%	94.29%
18	0.0490	0.1419	98.57%	94.29%
19	0.0634	0.1680	97.96%	95.71%
20	0.0582	0.1665	98.36%	95.71%

QNN Noisy

Metric	Value
Accuracy	76.43%
Precision	75.91%
Recall	76.43%
F1 Score	75.13%
Loss	0.6202
Training Time	34.39s
Train Loss	0.5517
Val Loss	0.6202
Train Accuracy	85.07%
Val Accuracy	77.14%
Train Size	489
Val Size	70
Test Size	140

QNN Noisy Confusion Matrix

Actual / Predicted	Class 0	Class 1
Class 0	83	9
Class 1	24	24

QNN Noisy Classification Report

Class	Precision	Recall	F1 Score	Support
0	77.57%	90.22%	83.42%	92.0000
1	72.73%	50.00%	59.26%	48.0000
macro avg	75.15%	70.11%	71.34%	140.0000
weighted avg	75.91%	76.43%	75.13%	140.0000
Accuracy			76.43%	

QNN Noisy Epoch History

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
1	0.6834	0.6638	68.92%	72.86%
2	0.6390	0.6468	77.71%	75.71%
3	0.6200	0.6368	81.60%	77.14%
4	0.6056	0.6309	82.41%	72.86%
5	0.5966	0.6271	83.03%	72.86%
6	0.5857	0.6225	84.25%	74.29%

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
7	0.5792	0.6200	83.64%	75.71%
8	0.5736	0.6195	84.25%	77.14%
9	0.5689	0.6197	84.46%	75.71%
10	0.5624	0.6191	84.46%	77.14%
11	0.5597	0.6188	84.87%	77.14%
12	0.5560	0.6192	84.66%	77.14%
13	0.5517	0.6202	85.07%	77.14%

QNN Mitigated

Metric	Value
Accuracy	82.14%
Precision	82.03%
Recall	82.14%
F1 Score	81.49%
Loss	0.5624
Training Time	2m 26.63s
Train Loss	0.4839
Val Loss	0.5624
Train Accuracy	86.50%
Val Accuracy	75.71%
Train Size	489
Val Size	70
Test Size	140

QNN Mitigated Confusion Matrix

Actual / Predicted	Class 0	Class 1
Class 0	85	7
Class 1	18	30

QNN Mitigated Classification Report

Class	Precision	Recall	F1 Score	Support
0	82.52%	92.39%	87.18%	92.0000
1	81.08%	62.50%	70.59%	48.0000
macro avg	81.80%	77.45%	78.88%	140.0000
weighted avg	82.03%	82.14%	81.49%	140.0000
Accuracy			82.14%	

QNN Mitigated Epoch History

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
1	0.6693	0.6229	60.53%	65.71%
2	0.6385	0.6050	64.21%	62.86%
3	0.6169	0.5909	65.03%	61.43%
4	0.5957	0.5797	79.55%	61.43%
5	0.5802	0.5732	80.78%	72.86%
6	0.5673	0.5678	82.82%	72.86%

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
7	0.5556	0.5634	84.25%	71.43%
8	0.5440	0.5606	84.25%	72.86%
9	0.5359	0.5581	84.87%	74.29%
10	0.5272	0.5573	85.28%	74.29%
11	0.5192	0.5558	85.28%	74.29%
12	0.5127	0.5553	85.07%	75.71%
13	0.5060	0.5555	85.69%	75.71%
14	0.5040	0.5557	85.69%	75.71%
15	0.4978	0.5563	86.30%	75.71%
16	0.4955	0.5570	86.30%	75.71%
17	0.4926	0.5584	85.69%	75.71%
18	0.4915	0.5595	85.89%	75.71%
19	0.4876	0.5608	86.30%	75.71%
20	0.4839	0.5624	86.50%	75.71%